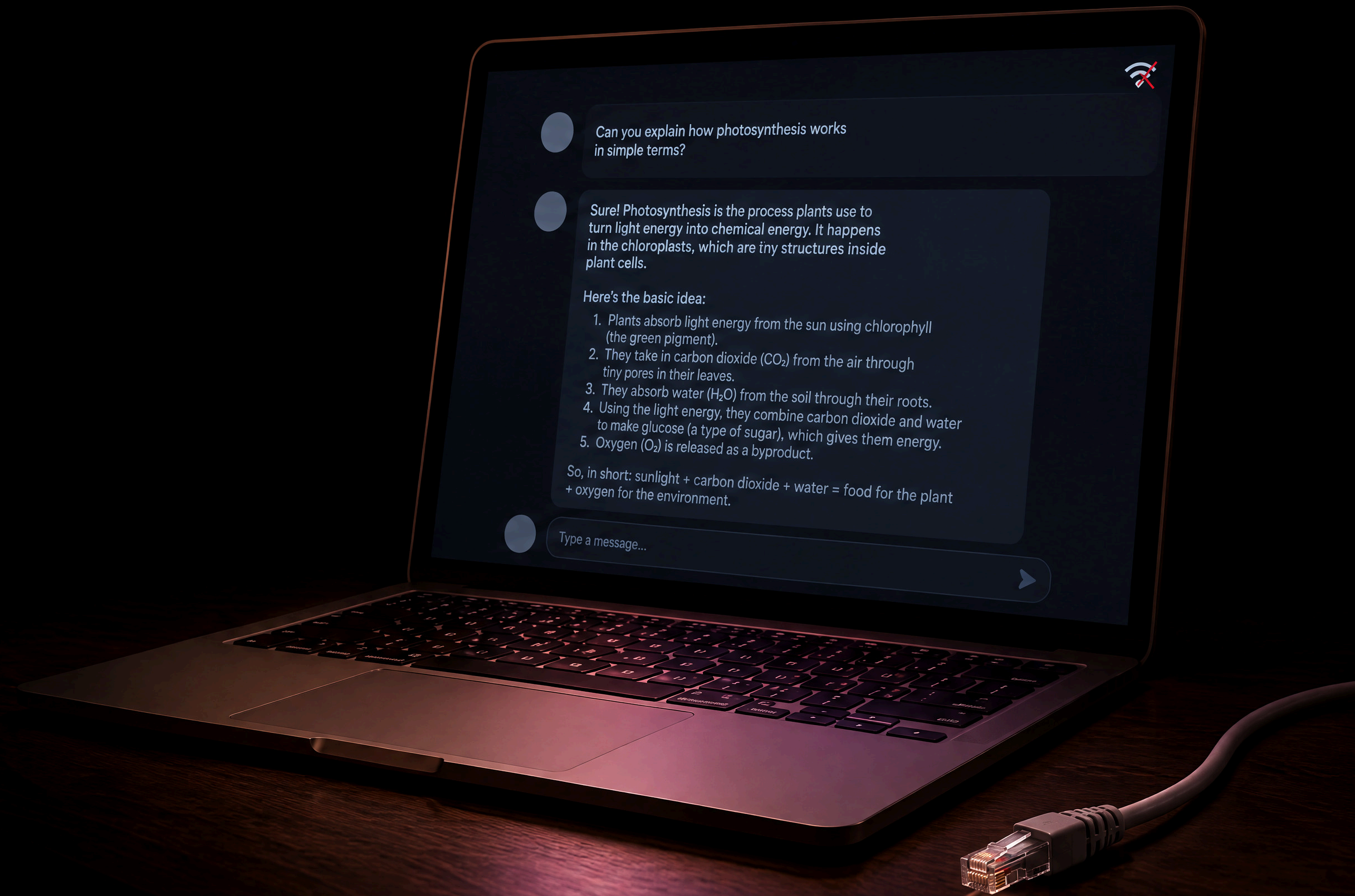


open models

artificial intelligence · prof. dr. nicolas meseth

- 1 what do you think?
- 2 offline
- 3 what a model is
- 4 open or closed

no internet,
and it still
answers.



part 1

what do you think?

when you chat with ChatGPT, where does the model run?

on my phone or laptop

in my browser

in a data centre

no idea



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on my phone or laptop

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when you chat with ChatGPT, where does the model run?

on my phone or laptop

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no idea



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how big is a language model that runs on a laptop?

a few megabytes

one to a few gigabytes

around 100 gigabytes

no idea



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a few megabytes



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around 100 gigabytes



no idea



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part 2

offline

step 1 · switch it off

turn off wi-fi.

step 1 · switch it off

turn off wi-fi.

load gemma 3 1b in LM Studio.

step 1 · switch it off

turn off wi-fi.

load gemma 3 1b in LM Studio.

ask it: “what does nitrogen fertiliser do for a wheat plant? two sentences.”

step 2 · find the file

LM Studio → my models → show in folder.

step 2 · find the file

LM Studio → my models → show in folder.

how big is the file, and what is it called?

step 3 · three questions, two models

| a field of 4.5 ha needs 140 kg of nitrogen per ha. how much is that in total?

step 3 · three questions, two models

| a field of 4.5 ha needs 140 kg of nitrogen per ha. how much is that in total?

| what is the capital of lower saxony, and what is it known for?

step 3 · three questions, two models

| a field of 4.5 ha needs 140 kg of nitrogen per ha. how much is that in total?

| what is the capital of lower saxony, and what is it known for?

| translate into german: “the harvest was delayed by rain.”

step 3 · three questions, two models

| a field of 4.5 ha needs 140 kg of nitrogen per ha. how much is that in total?

| what is the capital of lower saxony, and what is it known for?

| translate into german: “the harvest was delayed by rain.”

ask gemma 3 1b first, then gemma 3 4b, and note the tokens per second below each answer.

📶 quick poll

how many tokens per second did gemma 3 4b reach on your laptop?

fewer than 5

5 to 20

20 to 50

more than 50



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how many tokens per second did gemma 3 4b reach on your laptop?

fewer than 5



5 to 20



20 to 50



more than 50



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1b or 4b?

gemma 3 1b

gemma 3 4b

file

speed

quality



1b or 4b?

gemma 3 1b

gemma 3 4b

file

0.75 GB

2.4 GB

speed

quality

1b or 4b?

gemma 3 1b

gemma 3 4b

file

0.75 GB

2.4 GB

speed

your tokens/s?

your tokens/s?

quality

1b or 4b?

gemma 3 1b

gemma 3 4b

file

0.75 GB

2.4 GB

speed

your tokens/s?

your tokens/s?

quality

which answers held?

which answers held?

if a model like GPT-6.1 Sol lay on your laptop, how big would the file be?

about 5 gigabytes

about 50 gigabytes

about 500 gigabytes

several terabytes



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if a model like GPT-6.1 Sol lay on your laptop, how big would the file be?

about 5 gigabytes

☐

about 50 gigabytes

☐

about 500 gigabytes

☐

several terabytes

☐

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if a model like GPT-6.1 Sol lay on your laptop, how big would the file be?

about 5 gigabytes



about 50 gigabytes



about 500 gigabytes



several terabytes



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part 3

what a model is

a file full of numbers

gemma-3-4bgguf

2.4 GB

a file full of numbers

gemma-3-4bgguf

2.4 GB



0.0213	-1.4020	0.8710	-0.3361
-0.7745	0.0091	1.2203	0.5518
0.3302	-0.0687	-0.9154	0.1240
-1.1067	0.4432	0.0025	-0.6619

... and so on, about 4 billion times

a file full of numbers

gemma-3-4bgguf

2.4 GB

0.0213 -1.4020 0.8710 -0.3361
-0.7745 0.0091 1.2203 0.5518
0.3302 -0.0687 -0.9154 0.1240
-1.1067 0.4432 0.0025 -0.6619

... and so on, about 4 billion times

4 billion parameters, also called weights

how many values fit into a few bits?

1 bit → 2 values

how many values fit into a few bits?

1 bit → 2 values

2 bits → 4 values

how many values fit into a few bits?

1 bit	→	2 values
2 bits	→	4 values
4 bits	→	16 values

how many values fit into a few bits?

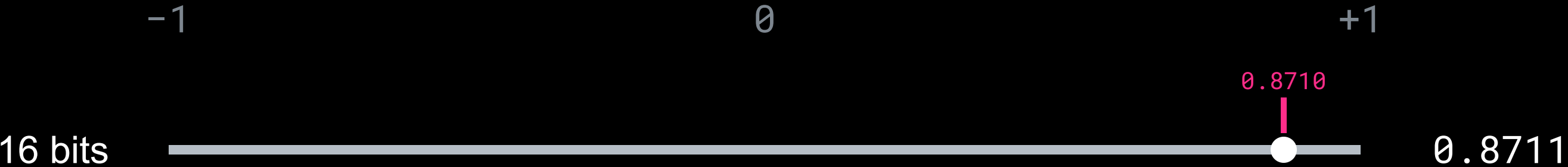
1 bit	→	2 values
2 bits	→	4 values
4 bits	→	16 values
16 bits	→	65,536 values

how many values fit into a few bits?

1 bit	→	2 values
2 bits	→	4 values
4 bits	→	16 values
16 bits	→	65,536 values

every extra bit doubles the number of values a weight can take.

rounding one weight

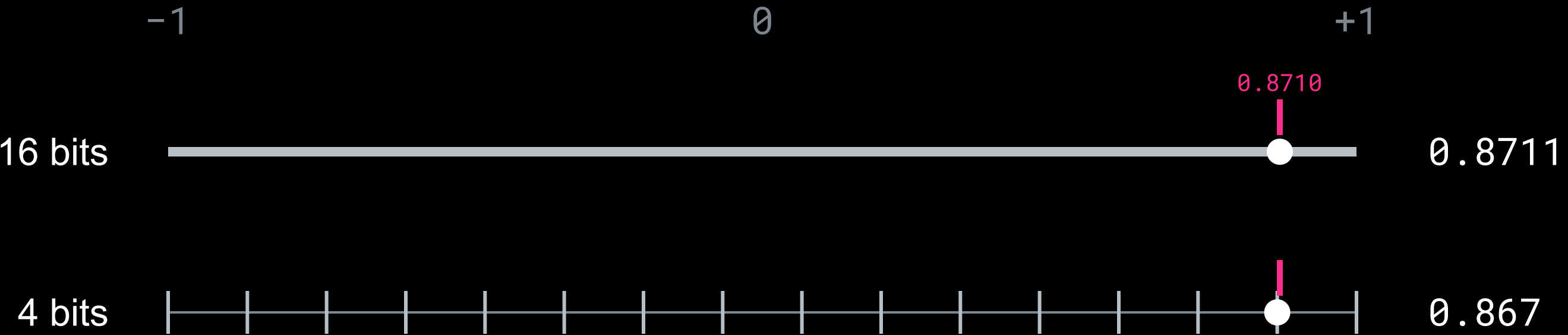


simplified: 16 evenly spaced values between -1 and $+1$



what a model is

rounding one weight

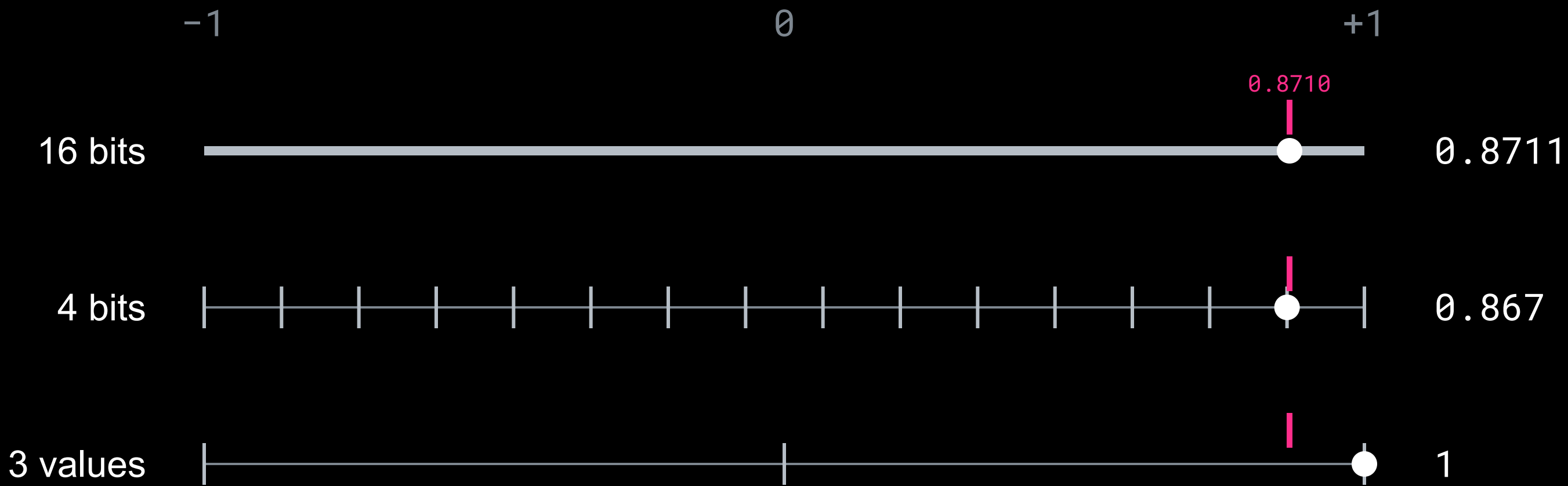


simplified: 16 evenly spaced values between -1 and +1



what a model is

rounding one weight

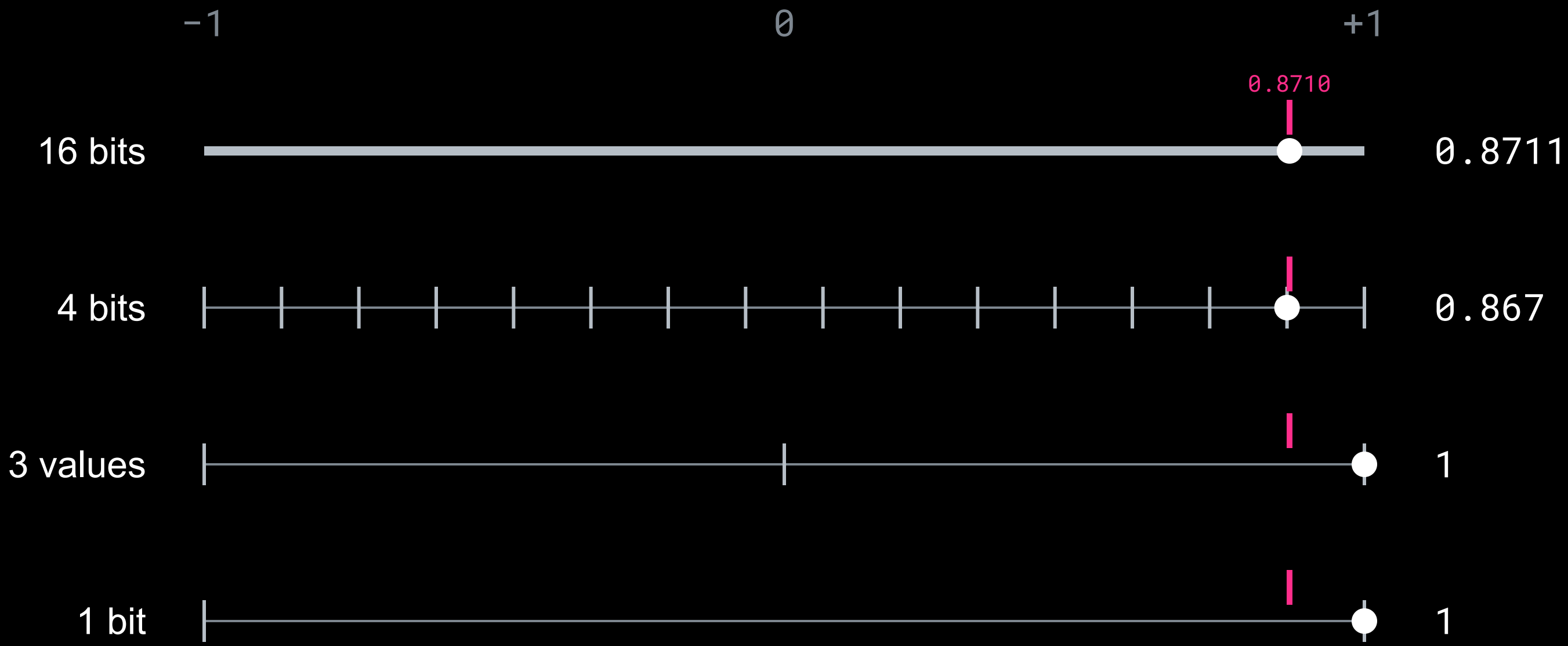


simplified: 16 evenly spaced values between -1 and +1



what a model is

rounding one weight



simplified: 16 evenly spaced values between -1 and +1



what a model is

four weights, rounded

16 bits

0.0213

-0.7744

0.3301

0.8711

four weights, rounded

16 bits	4 bits
0.0213	0.067
-0.7744	-0.733
0.3301	0.333
0.8711	0.867

four weights, rounded

16 bits	4 bits	3 values
0.0213	0.067	0
-0.7744	-0.733	-1
0.3301	0.333	0
0.8711	0.867	1

four weights, rounded

16 bits	4 bits	3 values	binary
0.0213	0.067	0	1
-0.7744	-0.733	-1	-1
0.3301	0.333	0	1
0.8711	0.867	1	1

four weights, rounded

16 bits	4 bits	3 values	binary
0.0213	0.067	0	1
-0.7744	-0.733	-1	-1
0.3301	0.333	0	1
0.8711	0.867	1	1

the coarser the rounding, the smaller the file, and the further the model drifts from what it learned.

why 2.4 GB?

4 billion numbers $\times$ 16 bits = 8 GB

why 2.4 GB?

4 billion numbers $\times$ 16 bits = 8 GB

4 billion numbers $\times$ ~4.5 bits = 2.4 GB

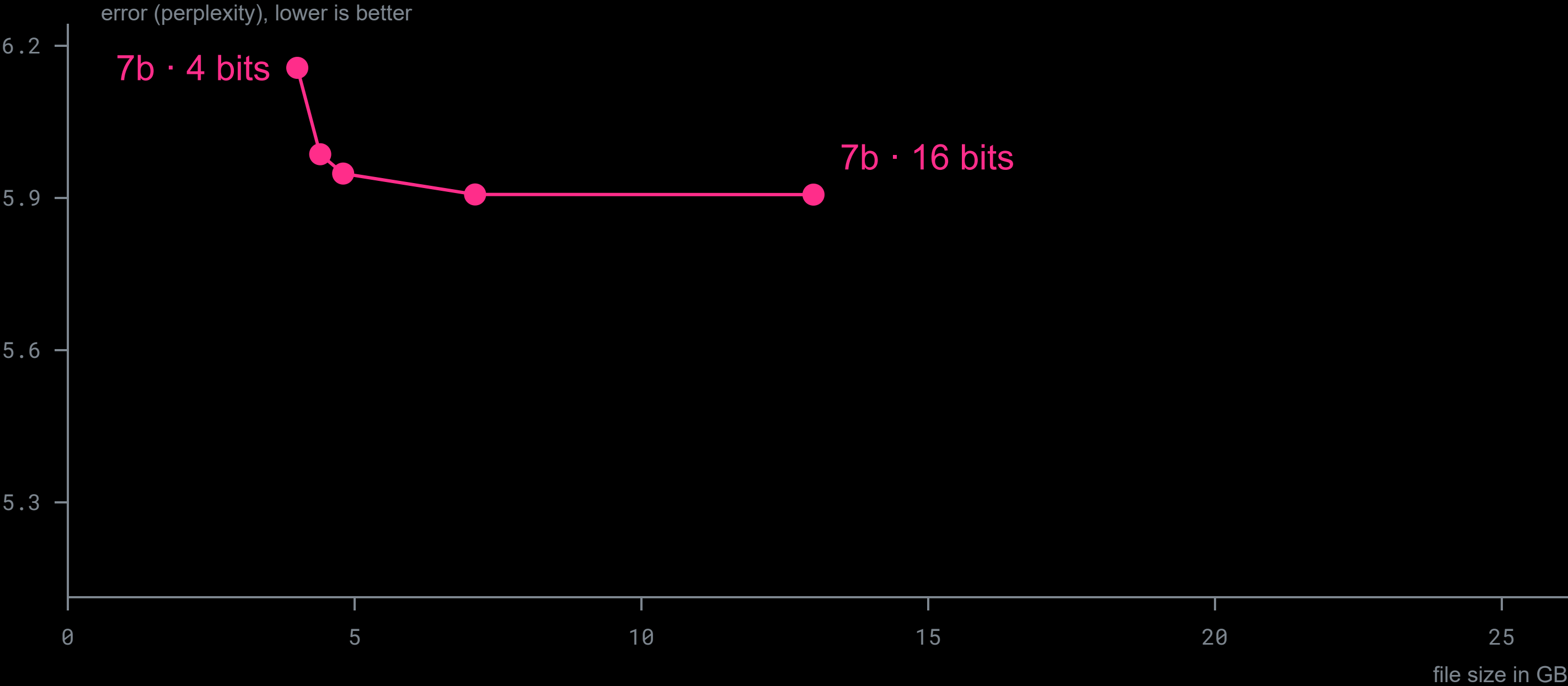
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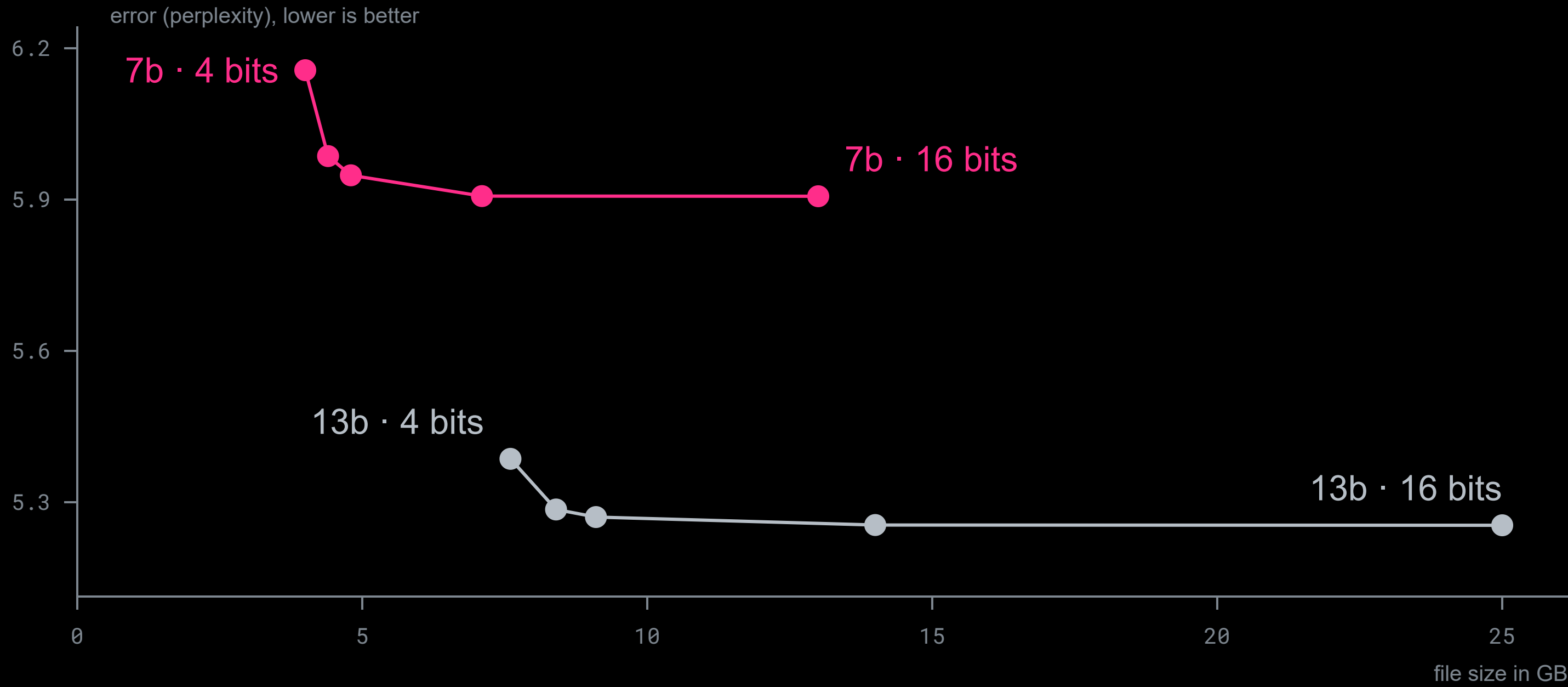
fewer bits per number: quantization.

rounding costs less than shrinking



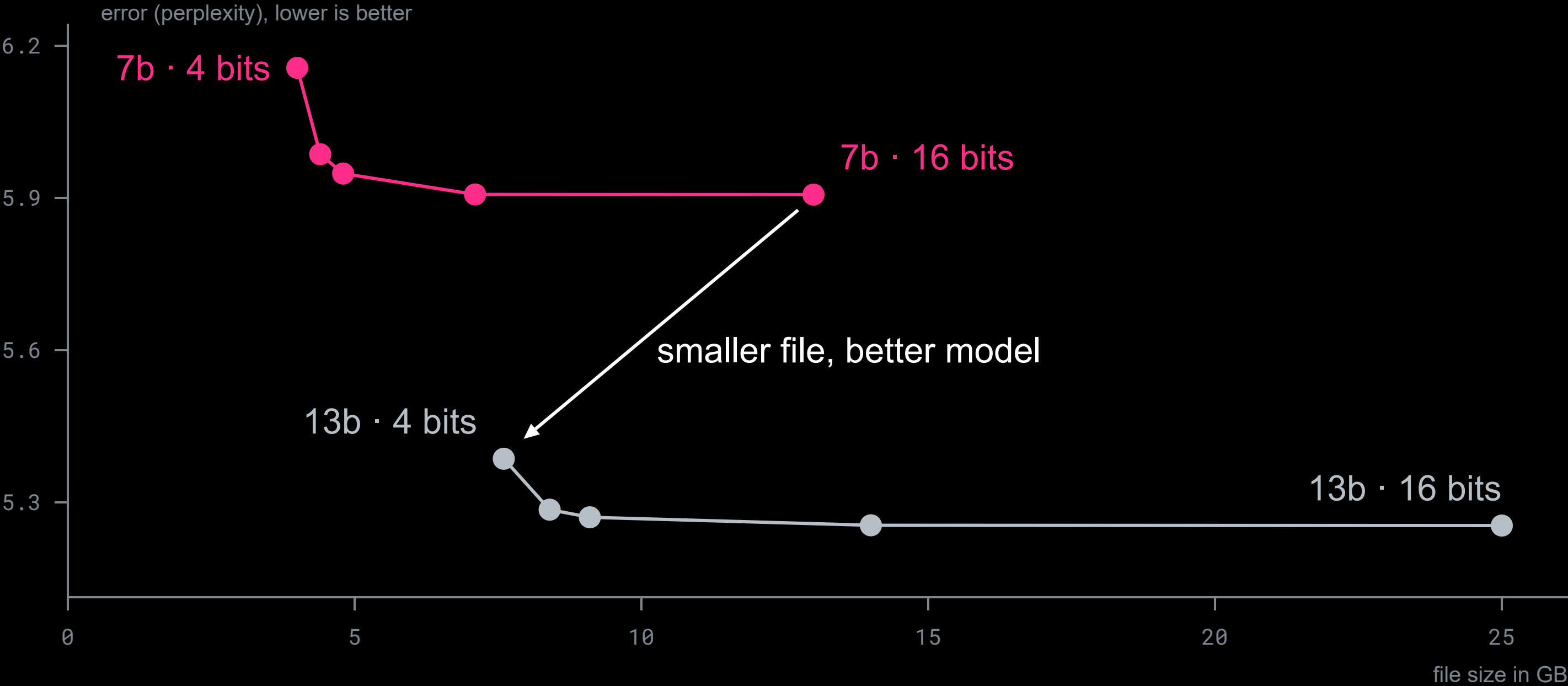
measured on LLaMA 7b and 13b: llama.cpp, 2023 · see also dettmers & zettlemoyer, the case for 4-bit precision, 2023

rounding costs less than shrinking



measured on LLaMA 7b and 13b: llama.cpp, 2023 · see also dettmers & zettlemoyer, the case for 4-bit precision, 2023

rounding costs less than shrinking



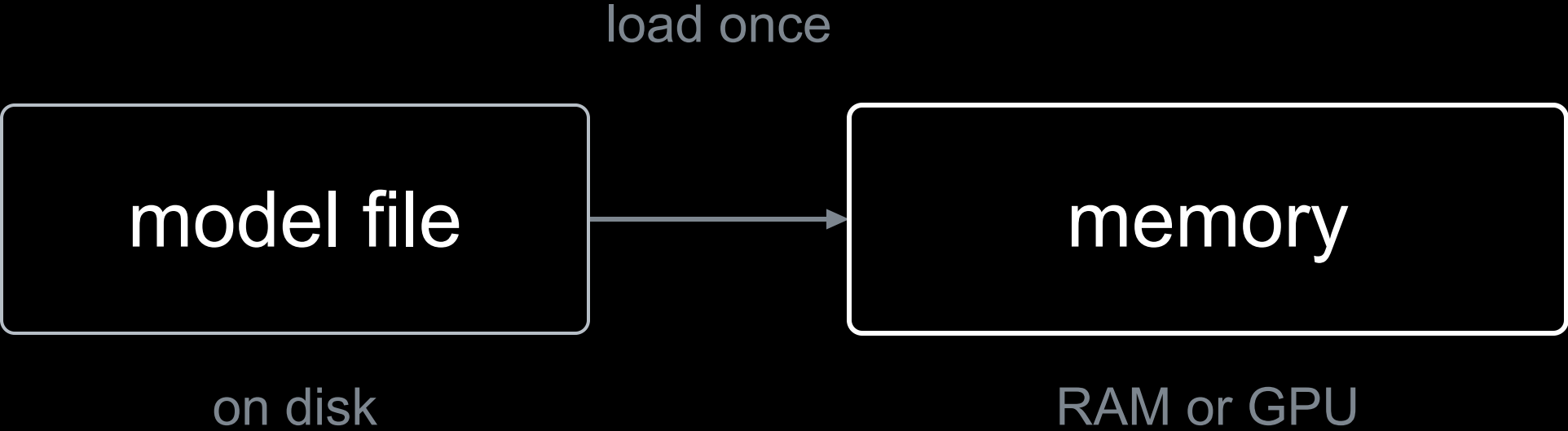
measured on LLaMA 7b and 13b: llama.cpp, 2023 · see also dettmers & zettlemoyer, the case for 4-bit precision, 2023

what happens when you press enter

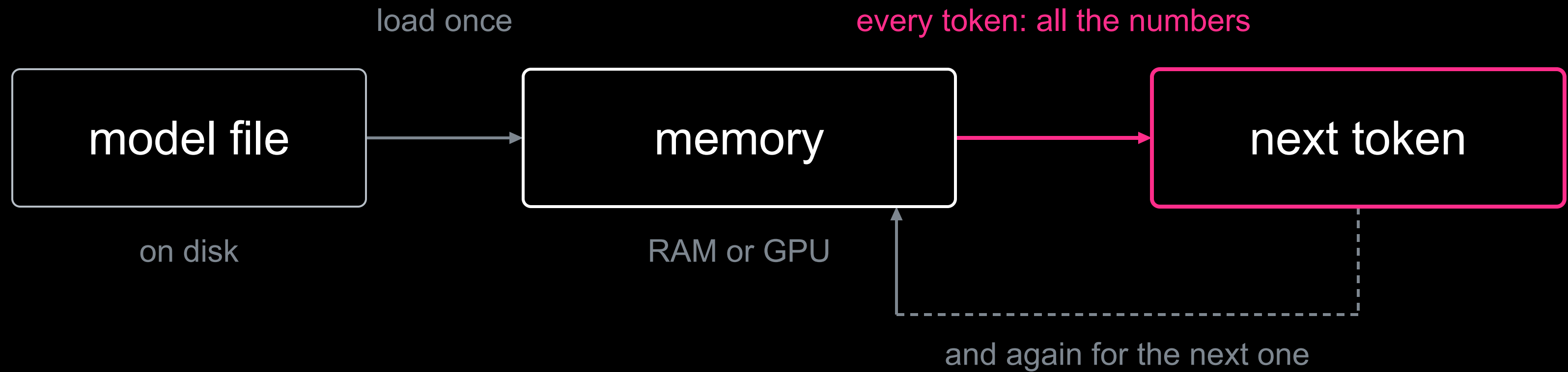
model file

on disk

what happens when you press enter

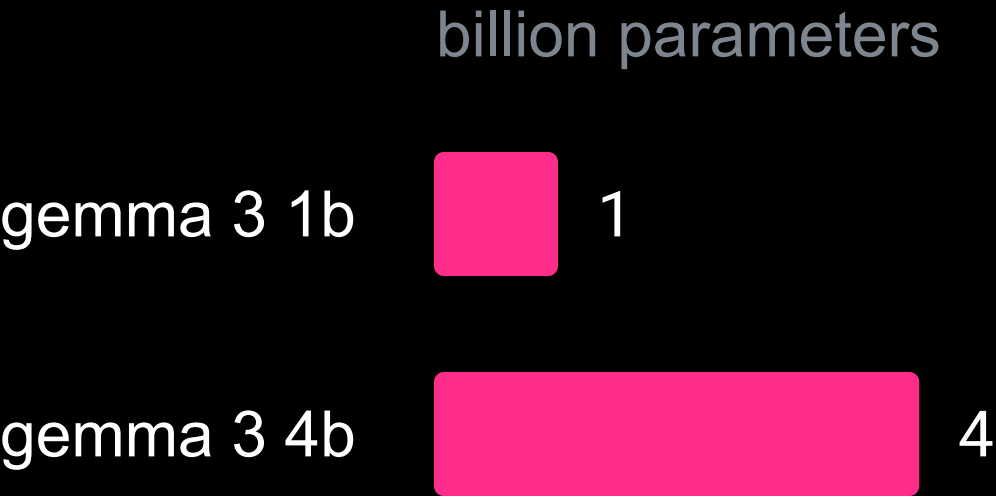


what happens when you press enter



bigger models are slower, because every token passes through more numbers.

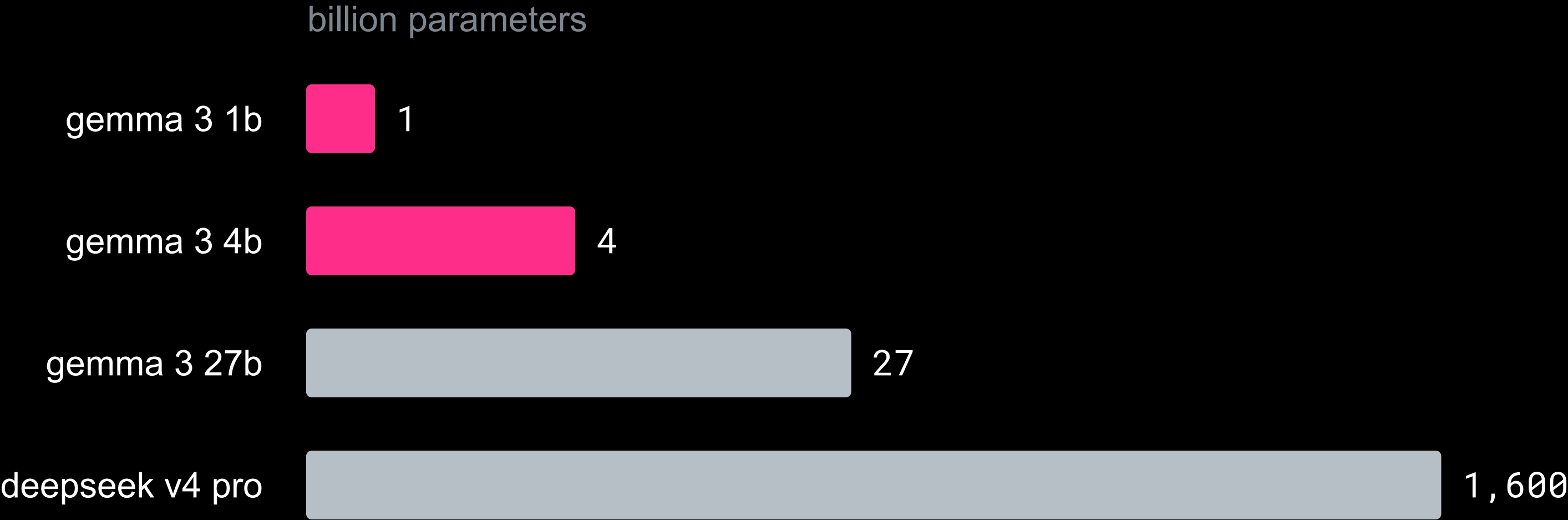
how big is big?



parameters as published by the providers; log scale

what a model is

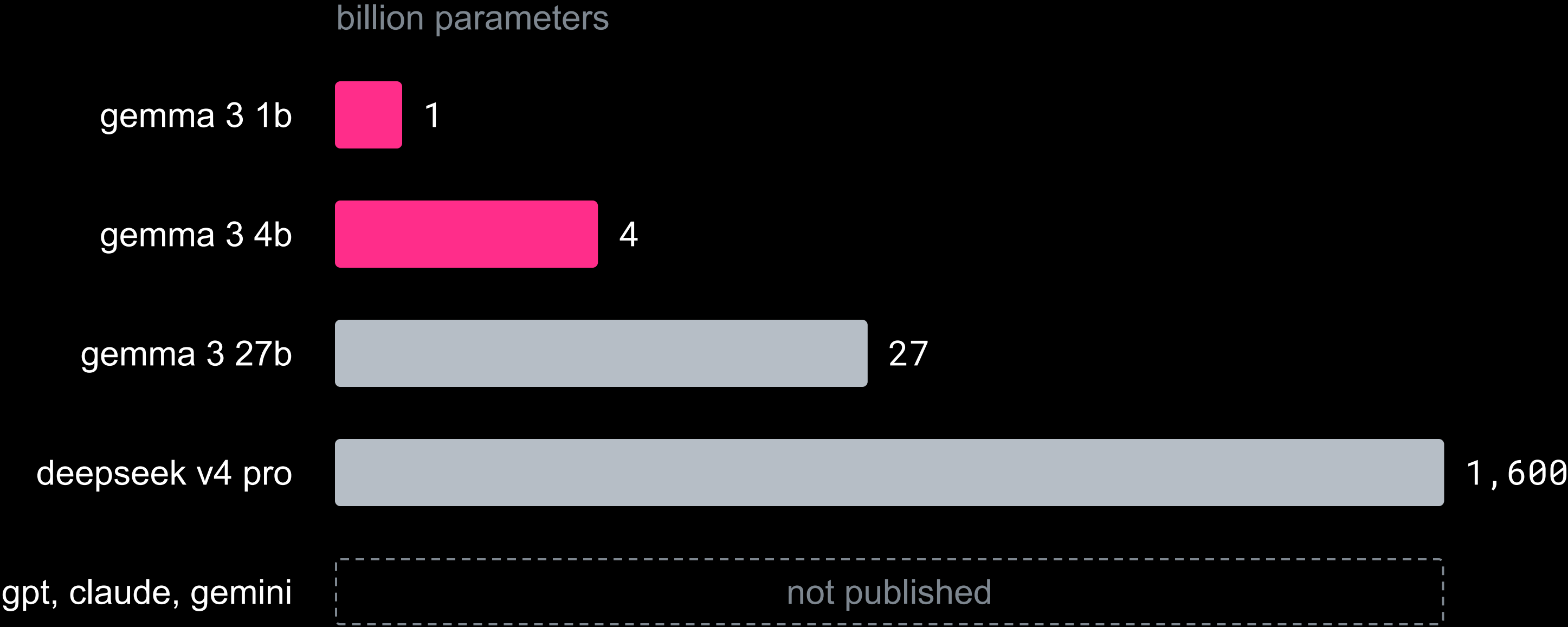
how big is big?



parameters as published by the providers; log scale

what a model is

how big is big?



parameters as published by the providers; log scale

local and cloud

the same harness can talk to a model on your laptop.

see also: the harness

what a model is

part 4

open or closed

open weights, closed weights

open

download it, run it,
change it.

gemma, qwen, llama, mistral, deepseek, gpt-oss

open weights, closed weights

open

download it, run it,
change it.

gemma, qwen, llama, mistral, deepseek, gpt-oss

closed

use it as a service.

gpt, claude, gemini

open weights, closed weights

open

download it, run it,
change it.

gemma, qwen, llama, mistral, deepseek, gpt-oss

closed

use it as a service.

gpt, claude, gemini

open weights is not the same as open source: the training data usually stays secret.

📱 quick poll

would you paste a farm's customer list into a cloud chatbot?

yes

only with a contract

only into a local model

never



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would you paste a farm's customer list into a cloud chatbot?

yes

only with a contract

only into a local model

never



lifi.uber.space/q

when local makes sense

your data must not leave the building.

when local makes sense

your data must not leave the building.

you need it offline, in a barn, on a tractor or on your phone.

on a phone: [google ai edge gallery](#), an app that runs small gemma models offline on android and ios

open or closed

when local makes sense

your data must not leave the building.

you need it offline, in a barn, on a tractor or on your phone.

you use it so much that tokens get expensive.

on a phone: [google ai edge gallery](#), an app that runs small gemma models offline on android and ios

open or closed

when local makes sense

your data must not leave the building.

you need it offline, in a barn, on a tractor or on your phone.

you use it so much that tokens get expensive.

and you accept a smaller, slower model in return.

on a phone: [google ai edge gallery](#), an app that runs small gemma models offline on android and ios

open or closed

a language model is **a file full of numbers** that runs on your own laptop without the internet; its **size** decides how good, how fast and how hungry it is.

discuss

what surprised you about the size, speed or quality of the local model?

discuss

what surprised you about the size, speed or quality of the local model?
which data would you trust a local model with, but not a cloud service?

discuss

what surprised you about the size, speed or quality of the local model?

which data would you trust a local model with, but not a cloud service?

when should a farm or a food company run its own model instead of renting one?